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Spectral bias — Section 4 (merged reply to related reviewer threads)

Q4 (Universality beyond mean–field):
I wonder how universal is spectral bias discovered in Section 4 for low–rank training? For example, if we do not consider network in the mean–field regime, do we still have such spectral bias?

Q5 (Qualitative discussion, two–point setting, high– vs low–frequency tilt):
While I acknowledge the spectral bias for low–rank network could be interesting, the discussions in Section 4.1 seems to be too qualitative, and the results in Section 4.2 seems to be on too simple setting (only two point in the datasets). Could the authors elaborate more on why the low–rank network lead to high–frequency bias, and fully trained network lead to low–frequency bias?

Answer (merged):

We thank the reviewers for connecting this discussion to the broader spectral–bias literature. In particular, Rahaman et al., "On the Spectral Bias of Neural "Networks (2019), arXiv:1806.08734, https://arxiv.org/abs/1806.08734, is an apt reference for the standard Fourier / low–frequency–first picture in sufficiently wide, kernel–like regimes.

On universality and training regimes: we will revise Section 4 to separate statements cleanly. In the NTK / lazy training regime, the effective bias is governed by the kernel spectrum and one expects the classical low–frequency–first behavior emphasized in that line of work. The empirical ""tilt toward comparatively higher–frequency structure that we highlight is observed in a richer feature–learning setting for our low–rank RF architecture; we do not claim that this behavior is universal across all scalings (including finite width outside mean–field parameterizations or lazy limits).

On exposition and scope: we agree Section 4 is currently qualitative and that the two–point analytic anchor is minimal. We will tone down claims, acknowledge the simplicity of the two–point construction, and add citations for the CPWL / compositional picture and Fourier heuristics beyond Rahaman et al. (e.g., Montúfar; Raghu et al., as already standard in this discussion).

On why low–rank maps might carry more high–frequency content (strictly heuristic, work in progress / under refinement for the manuscript): we will present one intuition alongside the boundary / region–count narrative, clearly labeled as not a theorem. In depth–L ReLU networks, one can view contributions through "sum over "paths (per–region linear parametrizations); shrinking the effective rank reduces how many independent hinge directions can co–activate across depth, which loosely relaxes constraints on how linear regions can tile input —spaceinformally, a less constrained CPWL can allow relatively more high–dimensional faces versus low–dimensional ones and thus alter how energy is distributed across spatial frequencies in a Fourier lens. This complements the shorter "fewer effectively independent hinge directions / boundary "geometry phrasing we will keep in the main text and will be marked explicitly as exploratory reasoning and ongoing review, not as a formal implication.

Concretely for the camera–ready, we will (i) separate NTK vs feature–learning claims, (ii) integrate the Rahaman et al. citation and related CPWL references, (iii) state the path / polytope / hinge–direction heuristic and the region–count argument only as intuition, and (iv) clarify that full–rank, fully trained networks are discussed relative to the usual low–frequency bias narrative, while our low–rank observations are regime–dependent and empirical.
